OSCE Past Exams: Exam Traps and Frequently Tested Points

Built from the past-exam OSCE compilations (2015 to 2025): Patient Safety, Neurology/Psychiatry, Nephrology, Hepatology, GI, Respiratory, Critical Care, Rheumatology, Endocrine, Infectious Diseases and Cardiovascular. ★ = station repeated in more than one exam year.

How examiners mark every station. Each answer key ends with the same checklist, so cover it in every case even if not asked: **(1) Goals** of therapy • **(2) Non-pharm** • **(3) First-line drug with dose, route, frequency, duration** • **(4) Special populations**: pregnancy, renal/dialysis, liver, HF • **(5) Interactions** • **(6) Counseling** • **(7) Monitoring**: efficacy and safety, with timing. A drug name without a dose and a monitoring plan loses marks.

PART A: THE MOST REPEATED STATIONS (★)

System	Station that keeps coming back	Years seen	What you must say
Cardio	AF on Flecainide + Diltiazem + Warfarin, then STEMI	2018–19, 2021, 2022–23, 2024	CHA2DS2-VASc/HAS-BLED, stop Flecainide (CAST), Diltiazem to β-blocker, Warfarin to Apixaban + Clopidogrel, Aspirin ≤1 week, OAC alone at 12 months
Cardio	STEMI / ACS with cath lab	2018–19, 2022–23, 2024, 2025	Aspirin 162–325 chewed, Ticagrelor 180, UFH 70–100 units/kg, PCI ≤90 min, Atorvastatin 80, contrast-AKI prevention, discharge list
Cardio	HFrEF on Amlodipine, EF <30%	2020, 2024	Stage C, NYHA class, 4 pillars with start/target doses, stop Amlodipine, Spironolactone checks (eGFR >30, K <5)
Cardio	Warfarin with nasal bleeding / INR high	2018–19, 2020, 2021, 2024	Hold, local pressure, oral Vit K 2.5–5 mg if minor; IV Vit K 5–10 + 4F-PCC if major; find the cause; recheck INR
Cardio	Pregnancy, Enoxaparin, low anti-Xa	2018–19, 2022–23	Peak 4–6 h after 3rd–4th dose; target 0.6–1.0; increase 10–25%; stop 24 h before delivery
Cardio	Stage 2 HTN + prediabetes smoker	2020, 2024	Two first-line agents, lifestyle numbers, NRT patch/gum doses, vaccines
Endocrine	DKA (staging and step-by-step)	2018–19, 2021, 2022–23, 2025	Fluids, K rules before insulin, 0.1 unit/kg/h, add D5 at 200, resolution criteria, SC overlap
Endocrine	Insulin TDD, basal-bolus, NPH/Mixtard switch	Every year 2018–2023	0.4–0.6 units/kg, 50/50, ICR 500/TDD, CF 1800/TDD, counseling and storage
Endocrine	Thyroid storm / myxedema / Graves	2018–19, 2020, 2021, 2022–23	PTU then iodine ≥1 h later, propranolol, hydrocortisone, acetaminophen not aspirin
ID	TB: RIPE, toxicities, INH hepatitis (difficult patient)	2018–19, 2021, 2022–23, 2025	Doses, Pyridoxine, stop rules (3× with symptoms, 5× without), rechallenge order R → H → Z
ID	Meningitis (± HTN emergency, gentamicin TDM)	2018–19, 2022–23	Ceftriaxone 2 g q12h + Vanco + Ampicillin (>50 y) + Dexamethasone before/with 1st dose
ID	HAP/VAP (on HD, valproate, citalopram)	2018–19, 2025	Vanco + Pip-tazo/Cefepime; no carbapenem with valproate; no linezolid with SSRI; 7 days
ID	HIV + TB	2020 (twice)	TB first; ART within 2 weeks if CD4 <50, within 8 weeks if ≥50; Dolutegravir 50 mg BID; TDF not TAF
GI	Crohn's + TPN calculations	2022–23, 2025	kcal, protein, dextrose g, GIR, lipids, K-phos; budesonide/infliximab plan
GI	H. pylori PUD	2020 (3 times)	Bismuth quadruple 14 days; hold PPI 2 weeks / antibiotics 4 weeks before testing

System	Station that keeps coming back	Years seen	What you must say
GI	Fulminant UC	2018–19, 2021	IV steroids, day-3 assessment, rescue infliximab/cyclosporine, avoid loperamide/opioids/NSAIDs
Hepatology	SBP / HRS / ascites / HE / HBV	2015, 2018–19, 2020, 2022–23	Ceftriaxone + albumin 1.5/1 g/kg; terlipressin + albumin; spironolactone:furosemide 100:40; lactulose
Nephrology	Contrast-induced AKI with urine indices	2020 (3 times)	NS 1 mL/kg/h 6–12 h before and after; hold metformin/NSAID/ACEi; FeNa; KDIGO stage
Critical care	Septic shock + vanco/genta TDM	2018–19, 2020	Hour-1 bundle, 30 mL/kg, NE → vasopressin, hydrocortisone 50 q6h, vanco load 25 mg/kg
Respiratory	COPD GOLD group / exacerbation; asthma step and inhaler counseling	2018–19, 2020, 2021, 2022–23, 2025	Group A/B/E choice, O2 88–92%, prednisone 40 × 5 d; MART; device technique
Respiratory	Smoking cessation	2018–19, 2021	Varenicline titration, bupropion contraindications, NRT doses
Rheum	RA on MTX (pregnancy planning, monitoring)	2018–19, 2020, 2022–23, 2025	MTX 15 → 25 mg weekly + folic acid; stop 3 months before conception; TB/HBV screen before biologic
Safety	Cefazolin anaphylaxis (angry patient); warfarin dispensing error	2020, 2021	Epinephrine IM first, apologize, report (SFDA + incident), allergy alert
Neuro	Stroke tPA eligibility; Carbamazepine + Warfarin	2018–19	Exclusions list; induction lowers INR → Levetiracetam

PART B: CARDIOVASCULAR TRAPS

Acute coronary syndrome

1. **Aspirin 162–325 mg chewed, non-enteric coated**, then 81 mg daily for life. Never write only 81 mg as the loading dose.
2. P2Y12 load: **Ticagrelor 180 mg** (then 90 mg BID) or **Prasugrel 60 mg** (only if age <75, weight ≥60 kg, no stroke/TIA) or **Clopidogrel 600 mg**. Clopidogrel **300 mg** is the dose with fibrinolysis.
3. Cath lab anticoagulant: **UFH 70–100 units/kg** (50–70 units/kg with GP IIb/IIIa) titrated to ACT; **Bivalirudin 0.75 mg/kg then 1.75 mg/kg/h** if HIT.
4. Time goals: first medical contact to device **≤90 min** (≤120 min if transferred); door-to-needle **≤30 min**; lysis only if PCI not possible within 120 min.
5. PCI beats lysis: TIMI 3 flow >90% vs 50–60%, less mortality, reinfarction and **intracranial hemorrhage (~1% with lytics)**.
6. Oxygen only if **SaO2 <90%**. Morphine 2–4 mg IV only for refractory pain: it **delays oral P2Y12 absorption** (use fentanyl or crush ticagrelor).
7. Oral β-blocker within 24 h **only if no HF, low output or shock**. Never IV β-blocker in a congested patient.
8. Statin: **Atorvastatin 80 mg** regardless of baseline LDL. Patient on fluvastatin/low-intensity statin → switch. LDL-C goal **<55 mg/dL** (reasonable) / <70; add **Ezetimibe 10 mg** if ≥70 on max statin, then PCSK9.
9. DAPT **≥12 months** after ACS (any stent or medical management); stable CAD with DES = 6 months. Ticagrelor counseling: dyspnea, aspirin dose ≤100 mg.
10. Discharge list examiners count: Aspirin, P2Y12, high-intensity statin, β-blocker, ACEi (EF ≤40%, HTN, DM, CKD), MRA (EF ≤40% + HF or DM), **SL NTG 0.4 mg q5min × 3 (call EMS if pain persists 5 min after the first dose)**, PPI if GI risk, **cardiac rehab** (often forgotten).
11. Nitroprusside in ACS = **coronary steal**; in renal impairment = thiocyanate. Use **IV nitroglycerin 10 mcg/min** for ischemia + HTN + pulmonary edema.

12. AKI after PCI: SCr rises within 24 h, peaks 2–5 days. Hold ACEi/ARB/MRA/metformin/NSAIDs; no change for aspirin, ticagrelor, atorvastatin. Enoxaparin q24h if CrCl <30; eptifibatide 50% if CrCl <50; rosuvastatin max 10 mg if CrCl <30.
13. Contrast AKI prevention: isotonic saline; **bicarbonate and NAC add no benefit (PRESERVE)**; hold metformin 48 h; no theophylline, no prophylactic dialysis.
14. Depression after MI: **Sertraline** (SADHART). Venlafaxine raises BP/HR → **taper** (75 → 37.5 mg for 1–2 weeks), never stop abruptly. SSRI + DAPT → add PPI (pantoprazole).
15. Statin stop rules: **CK >10× ULN with symptoms** or rhabdomyolysis; **persistent ALT >3× ULN**. Mild myalgia → hold, rechallenge lower dose or another statin.

Atrial fibrillation (the most repeated cardio station)

1. CHA2DS2-VASc: age ≥75 = **2**, stroke/TIA = **2**, age 65–74 = 1, female = 1, vascular (MI) = 1. After the STEMI the score goes **up by 1**. Example: 69 y + HTN + DM = 3; after MI = 4; woman = +1.
2. OAC if score **≥2 men / ≥3 women**. Age 55 scores 0 (points start at 65). CHADS2 counts age only at ≥75.
3. HAS-BLED ≥3 = high bleeding risk, **not a reason to withhold OAC**: fix BP, NSAIDs, alcohol, labile INR. State your assumptions for missing data.
4. **Flecainide/Propafenone are contraindicated after MI or structural heart disease (CAST)**. Replace with Amiodarone, Dofetilide (or Sotalol if EF preserved).
5. **Diltiazem/Verapamil are harmful in HFrEF** and lose post-MI benefit → switch to Metoprolol succinate / Bisoprolol.
6. AF + PCI: **DOAC (Apixaban) + Clopidogrel**; Aspirin only peri-PCI up to 1 week (max 30 days); OAC alone after 12 months. Never triple therapy long term.
7. Apixaban **2.5 mg BID only if ≥2 of: age ≥80, weight ≤60 kg, SCr ≥1.5**. The 57-kg, 69-year-old woman with CrCl 95 has only 1 criterion → **5 mg BID**.
8. Warfarin → DOAC: stop warfarin, start Apixaban/Dabigatran when **INR <2**, Rivaroxaban when **INR <3**, Edoxaban when **INR ≤2.5**. No bridging.
9. Once-daily DOAC for CrCl 15–50: **Rivaroxaban 15 mg with the evening meal** or Edoxaban 30 mg. On dialysis → **Apixaban 5 mg BID** (2.5 if ≥80 y or ≤60 kg); avoid dabigatran/edoxaban.
10. Rate target: lenient **<110** if asymptomatic; **<80** if symptomatic or HF. HR 95 with fatigue = symptomatic → tighten.
11. Mechanical valve or moderate–severe mitral stenosis → **Warfarin only** (RE-ALIGN). Mitral mechanical INR **2.5–3.5**. Offer home INR self-testing, not a DOAC.
12. Cardioversion: AF ≥48 h → ≥3 weeks OAC or TEE first, continue ≥4 weeks after. Pill-in-the-pocket **Flecainide 300 mg** with an AV nodal blocker (no structural disease).
13. Stroke + AF: start OAC within 48 h for minor/moderate stroke, day 6–7 for major (ELAN). **No aspirin on top** unless a separate indication (recent ACS/stent).
14. Amiodarone interactions: **Warfarin ↓30–50%**, **Digoxin ↓30–50%**, **Simvastatin max 20 mg** (Lovastatin 40); Atorvastatin/Rosuvastatin need no cap. Two class III agents together (Amiodarone + Dofetilide) → stop Dofetilide.
15. Amiodarone monitoring: LFT + TFT every 6 months, CXR yearly, PFT/DLCO baseline and if symptoms, eye exam, ECG; photosensitivity, blue-gray skin.
16. Amiodarone-induced thyrotoxicosis: Type 1 (iodine load) → **Methimazole 40 mg**; Type 2 (thyroiditis) → **Prednisone 40 mg**; mixed → both. Amiodarone is 37% iodine, half-life ~58 days.
17. QT patient (Dofetilide, QTc prolonged) needs an antibiotic → **Doxycycline**, not azithromycin/quinolone. Keep K >4, Mg >2; hold Dofetilide if QTc >500 ms.
18. Fluoxetine + Warfarin (CYP2C9 → INR up), + Amiodarone (QT), + Insulin (hypoglycemia). Prefer Sertraline.

Heart failure

1. Stage vs class: **ACC stage C** = structural disease + symptoms; **NYHA II** = ordinary activity causes symptoms; **NYHA III** = symptoms with minimal activity. Answer both 'number and letter'.
2. EF: HFrEF ≤40%, HFmrEF 41–49%, HFpEF ≥50%, HFimpEF = previously ≤40% now >40%.
3. 4 pillars: ARNI, evidence-based β-blocker (**only Bisoprolol, Carvedilol, Metoprolol succinate**), MRA, SGLT2i. Metoprolol **tartrate** is not GDMT.

4. **ACEi → ARNI: 36-hour washout**. Start dose: 24/26 mg BID if low-dose ACEi (≤10 mg enalapril-equivalent per day); **49/51 mg BID** if standard dose. Target 97/103 BID.
5. Doctor wants ACEi → Losartan: **disagree** (ARB not superior); switch to ARB only for cough/intolerance; better move is ARNI.
6. Start doses: Bisoprolol 1.25 → 10; Carvedilol 3.125 BID → 25 BID (50 BID if >85 kg); Metoprolol succinate 12.5–25 → 200; Spironolactone 12.5–25; Dapagliflozin/Empagliflozin 10.
7. Spironolactone checks: **eGFR >30, K <5.0, SCr ≤2.5 men / ≤2.0 women**; monitor K/SCr at **~1 week, 1 month, 6 months**; K >5.5 → halve or stop. K 5.0 after 2 months = acceptable, recheck.
8. **Stop Amlodipine** in HFrEF: no benefit, edema mimics congestion, steals BP room from GDMT. Avoid NSAIDs, TZDs, non-DHP CCBs.
9. Start low and titrate: ACEi/ARB double q1–4 weeks, ARNI q2–4 weeks, β-blocker q1–2 weeks **only when euvolemic**. Never up-titrate β-blocker in an overloaded patient.
10. Mortality-reducing classes: ARNI/ACEi/ARB, β-blocker (~35%), MRA (~30%, RALES), SGLT2i, Hydralazine/ISDN (African American). **Digoxin, Ivabradine, Vericiguat, loop diuretics do not reduce mortality.** Digoxin level 0.5–0.9 ng/mL.
11. Ivabradine needs **sinus rhythm** and HR ≥70 on max β-blocker → not in AF.
12. ADHF warm and wet: IV furosemide **1–2.5× home oral dose** (20 mg IV = 40 mg PO). Response: UOP >100–150 mL/h or spot urine Na >50–70 at 2 h; else double. Resistance → Metolazone 2.5–5 mg 30 min before loop, Acetazolamide 500 mg IV (ADVOR), or infusion.
13. ADHF with K 5.6: hold ACEi, no MRA until K <5. Penicillin-allergic pyelonephritis → Levofloxacin 750 mg × 5–7 days (check QTc).
14. ICD if EF ≤35% after ≥3 months GDMT (≥40 days post-MI). Repeat echo 3–6 months after GDMT optimization. NT-proBNP (not BNP) on ARNI.
15. Echo answer list: LVEF, LV size/dilation, LVH, regional wall motion, valves (MR), diastolic function, RV function, LV thrombus.

Hypertension and dyslipidemia

1. Stage 2 HTN (≥140/90) → **two first-line agents from different classes**, ideally single pill (e.g. Lisinopril 10 + Amlodipine 5). Goal **<130/80**.
2. Lifestyle numbers: weight −1 mm Hg/kg; DASH −8 to −14; sodium <2300 (ideally 1500) −6 to −8; dietary K 3500–5000 −6; exercise −4 to −10; alcohol ≤2/≤1 drinks.
3. DM + HTN + albuminuria → ACEi/ARB first. SCr up to 30% rise acceptable; check K/SCr 1–2 weeks.
4. HTN emergency: lower MAP **≤25% in the first hour**, then 160/100–110 over 2–6 h, then normal over 24–48 h. Labetalol 20 mg IV → 20–80 q10 min (max 300) or Nicardipine 5 mg/h. NSAIDs (ibuprofen for headache) can trigger it.
5. Statin intensity: 10-year risk 7.5–19.9% (e.g. 14%) → **moderate** (Atorva 10–20, Rosuva 5–10); ≥20% or ASCVD → **high** (Atorva 40–80, Rosuva 20–40). DM 40–75 → at least moderate; high if risk factors.
6. PCSK9 inhibitors (Evolocumab 140 q2w / 420 monthly; Alirocumab 75–150 q2w; Inclisiran 284 mg at 0, 3 months then q6 months) only for **ASCVD or FH** above goal on statin + ezetimibe. Not in primary prevention at 14% risk.
7. Aspirin for primary prevention: **No**. Secondary prevention: Yes.
8. Statin monitoring: lipids 4–12 weeks, LFT baseline only, CK only with symptoms. Short half-life statins (simva, lova, prava) in the evening. Diltiazem/Verapamil → Simvastatin ≤10 mg.
9. TG ≥500 → fibrate/omega-3 to prevent pancreatitis; Icosapent ethyl 2 g BID if TG 135–499 with ASCVD on statin. Gemfibrozil + statin = rhabdomyolysis.

VTE and anticoagulation

1. Enoxaparin 1 mg/kg q12h (62.4 kg → **60 mg q12h**) or 1.5 mg/kg daily; **q24h if CrCl <30**. Massive PE with planned lysis → **UFH 80 units/kg + 18 units/kg/h**.
2. Massive PE (SBP <90): **Alteplase 100 mg over 2 h**; hold heparin during infusion, resume when aPTT <2× control.
3. Warfarin overlap: **≥5 days AND INR ≥2 for 24 h**. Early high INR on day 3 = factor VII effect, not full anticoagulation → keep heparin.
4. INR adjustments: slightly out (≤0.5) → may keep dose; INR 3.7 → hold/skip one dose, reduce weekly 5–20%; INR 1.3 after recent PE → **bridge with enoxaparin** + increase 10–20%; INR 1.8 → +5–10%.

5. Carbamazepine (and Rifampin, Phenytoin) induce warfarin → INR falls over 1–4 weeks → choose **Levetiracetam 500 mg BID**. Azathioprine also lowers INR; when stopped, INR rises.
6. Antibiotic + infection on warfarin (INR 4.5, no bleeding): hold 1–2 doses, resume 5–20% lower, recheck after the course.
7. Bleeding on warfarin: minor → hold, local pressure 10–15 min, oral Vit K 2.5–5 mg if INR ≥4.5; major → **Vit K 5–10 mg slow IV + 4F-PCC** (25/35/50 units/kg for INR 2–<4 / 4–6 / >6). FFP only if no PCC.
8. Reversal on DOAC: **Idarucizumab 5 g** for dabigatran; 4F-PCC or Andexanet for apixaban/rivaroxaban.
9. Cancer VTE: DOAC (Apixaban/Edoxaban/Rivaroxaban) or LMWH; **prefer LMWH or Apixaban with luminal GI/GU cancer**. Continue while cancer is active; second VTE → indefinite. Extended phase: Apixaban 2.5 BID / Rivaroxaban 10 mg after 6 months.
10. Prophylaxis: Enoxaparin 40 mg daily (BMI ≥40: 40 mg BID); extend 4 weeks after major abdominal/pelvic cancer surgery.
11. **APS (triple positive) → Warfarin, not DOAC.** Send thrombophilia tests before heparin when possible (protein C/S/AT unreliable acutely).
12. Pregnancy anti-Xa: peak **4–6 h after the 3rd–4th dose**; treatment q12h target **0.6–1.0** (0.25 = low → increase 10–25%, repeat after 3rd–4th new dose). Stop 24 h before delivery/neuraxial; restart 4–6 h after vaginal, 6–12 h after cesarean; ≥6 weeks postpartum.
13. Estrogen-containing contraceptives after VTE are contraindicated → IUD/progestin; Warfarin is teratogenic (weeks 6–12) but safe in breastfeeding.

Other cardio

1. Bradycardia not responding to atropine (1 mg q3–5 min, max 3 mg): **pacing, Dopamine 5–20 mcg/kg/min or Epinephrine 2–10 mcg/min**. β-blocker toxicity → glucagon 3–5 mg; digoxin toxicity → Fab.
2. Multi-drug counseling station (Amiodarone, Warfarin, Digoxin, Simvastatin, Carvedilol): open with introduction and what the patient knows, check every interaction, close with **teach-back**.
3. NRT patch: **>10 cig/day 21 mg × 6 w → 14 × 2 → 7 × 2**; ≤10 cig/day start 14 mg. Gum **4 mg if first cigarette within 30 min** of waking; ≥9 pieces/day, max 24, chew-and-park.
4. Vaccines when none for 20 years: Tdap then Td q10 years, influenza, PCV20/21, COVID-19, HBV (19–59), zoster ≥50, RSV per age/risk.

PART C: ENDOCRINE TRAPS

DKA / HHS

1. Diagnosis: glucose >250 (or known DM), pH <7.3, HCO3 <18, ketones; AG = Na − (Cl + HCO3). Severity by pH: mild 7.25–7.30, moderate 7.00–7.24, severe <7.00.
2. Order: **fluids first** (NS 15–20 mL/kg in the 1st hour), then **check K before insulin**.
3. **K <3.3: hold insulin**, give K 20–30 mEq/h; **3.3–5.2**: add 20–30 mEq per liter; **>5.2**: no K, recheck q2h.
4. Insulin regular IV 0.1 unit/kg bolus + 0.1 unit/kg/h (or 0.14 units/kg/h without bolus); glucose should fall 50–75 mg/dL/h.
5. **Add D5 when glucose <200** (HHS: <300) and reduce insulin to 0.02–0.05 unit/kg/h. Do not stop insulin to fix glucose: insulin is for the ketones.
6. Bicarbonate only if **pH <7.0**. Phosphate only if <1 mg/dL or cardiac/respiratory compromise.
7. Resolution: glucose <200 + 2 of (HCO3 ≥15, venous pH >7.3, AG ≤12). **Overlap SC basal 1–2 h before stopping the drip.**
8. Transition (insulin-naive): 0.5–0.8 units/kg/day, half basal half bolus (70 kg × 0.5 = 35 → glargine ~17 + aspart 6 TID).
9. Discharge after DKA: avoid SGLT2i; stop glibenclamide in CKD; metformin max 1000 mg at eGFR 30–44, stop <30.

Insulin calculations and counseling

1. T1DM TDD **0.4–0.6 units/kg**; **50% basal / 50% bolus** divided over 3 meals.
2. **ICR = 500/TDD** (g carbohydrate per unit); **CF = 1800/TDD** (mg/dL drop per unit; 100/TDD in mmol/L).
3. NPH/Mixtard to basal-bolus: add up all current units = TDD → 50:50; **reduce 10–20% if hypoglycemia** or poor intake. Mixtard 70/30: 70% of each dose is NPH.

4. T2DM basal start: **10 units or 0.1–0.2 units/kg**, titrate +2 units every 3 days to fasting target 80–130.
5. Insulin indicated at diagnosis if **A1C >10%**, glucose ≥300, or catabolic symptoms.
6. Pattern from glucose log: adjust the insulin that covers the high period by 10–20% or 2 units; fasting highs → basal; post-meal highs → that meal's bolus.
7. Counseling: storage in fridge, in-use pen at room temperature **28 days**; rotate sites; rapid-acting 0–15 min before meals; **15-15 rule** for hypoglycemia; glucagon for family; sick-day rules.
8. Dialysis: reduce insulin 25–50%; linagliptin needs no adjustment; A1C unreliable in HD/anemia.

Type 2 DM drug choice

1. ASCVD/HF/CKD → GLP-1 RA or SGLT2i regardless of A1C. HF/cardiomyopathy → **avoid TZD, saxagliptin, alogliptin**.
2. Before semaglutide: personal/family **MTC or MEN2**, pancreatitis history, retinopathy, gastroparesis, pregnancy. Titration 0.25 → 0.5 → 1 (→ 2) mg weekly every 4 weeks; oral 3 → 7 → 14 mg.
3. Pancreatitis history → avoid GLP-1 RA and DPP-4i. Recurrent UTI/genital infections → avoid SGLT2i.
4. Albuminuria A3 (>300 mg/g): max-dose ACEi/ARB + SGLT2i + **Finerenone**. Statin for all DM 40–75.
5. Pregnancy: switch orals to **insulin**; targets fasting <95, 1 h <140, 2 h <120; low-dose aspirin 81–162 from 12 weeks; stop ACEi/ARB/statin/GLP-1/SGLT2.
6. Elderly/frail A1C <8%. Neuropathy: duloxetine, pregabalin, gabapentin.

Thyroid

1. Thyroid storm order: **PTU 500–1000 mg load then 250 mg q4h → iodine at least 1 h after** the thionamide → **Propranolol 60–80 mg q4–6h → Hydrocortisone 100 mg q8h** → supportive.
2. Fever in storm: **acetaminophen, not aspirin** (aspirin displaces T4 from TBG).
3. Myxedema coma: **hydrocortisone first**, then **LT4 200–400 mcg IV** load, then 50–100 mcg daily.
4. Graves: Methimazole 10–30 mg for 12–18 months; **PTU in the 1st trimester**, then switch. Agranulocytosis: sore throat + fever → stop and check CBC.
5. Levothyroxine under-replaced: first check **adherence and administration** (empty stomach, 30–60 min before breakfast, 4 h from Ca/Fe/PPI), then +12.5–25 mcg; TSH in 6–8 weeks.
6. Pregnancy: increase LT4 **25–30% (2 extra tablets/week)** once pregnant; TSH <2.5; check q4 weeks in the first half.
7. Rifampin increases levothyroxine requirement → recheck TSH 4–6 weeks.

PART D: INFECTIOUS DISEASES TRAPS

Tuberculosis (asked every year)

1. Doses: **INH 5 mg/kg (300)**, **RIF 10 mg/kg (600)**, **PZA 20–25 mg/kg (~1500)**, **EMB 15–20 mg/kg (~1200)**, **Pyridoxine 25–50 mg**. 2 months RIPE + 4 months INH/RIF = 6 months.
2. Extend to 9 months if cavitation + culture positive at 2 months. CNS 9–12 months; bone 6–9 months. Duration counts **doses taken**, not calendar time.
3. Tablet count (56–75 kg): INH 300 × 1, RIF 300 × 2, PZA 500 × 3, EMB 400 × 3 ≈ **9 tablets** + B6; fixed-dose 4-in-1: 55–70 kg = **4 tablets**.
4. **Hepatotoxicity stop rule: ALT ≥3× ULN with symptoms OR ≥5× ULN without.** At 4× ULN asymptomatic → continue with weekly LFT.
5. Rechallenge when ALT <2× ULN: **Rifampin first → INH after ~1 week → PZA after another week**. PZA is the one usually left out (then extend to 9 months).
6. Bridge while holding: Ethambutol + Levofloxacin/Moxifloxacin (± aminoglycoside).
7. Latent TB INH hepatitis: stop INH, **do not rechallenge**; complete with **Rifampin 4 months (4R)**. 3HP and 3HR contain INH → avoid.
8. Difficult patient who missed the visit and wants a refill: **do not dispense** before checking symptoms and LFTs; no blame; limit acetaminophen ≤2 g/day; report the ADR.
9. Vision change → **Ethambutol** (stop, ophthalmology). Gout → **PZA** (do not reduce the dose; colchicine/prednisone). Neuropathy → INH (B6). Orange fluids → RIF.
10. Renal: **INH and RIF no adjustment**; PZA 25 mg/kg and EMB 15–25 mg/kg **three times weekly after HD** if CrCl <30.

11. Rifampin interactions: warfarin, DOACs, tacrolimus, OCs, steroids, levothyroxine, azoles. **Voriconazole + Rifampin = contraindicated** (~95% fall) → **liposomal Amphotericin B 3–5 mg/kg**.
12. Amphotericin monitoring: infusion reactions (premedicate acetaminophen + diphenhydramine ± hydrocortisone; meperidine for rigors), **NS 500–1000 mL before each dose**, SCr, K and Mg wasting.
13. Isolation off when: ≥2 weeks effective therapy + clinical improvement + **3 negative AFB smears ≥8 h apart** (one early morning).
14. HIV + TB: **TB first**; ART within **2 weeks if CD4 <50**, within **8 weeks if ≥50** (CD4 82 → 8 weeks). Reason: IRIS, overlapping toxicity, pill burden.
15. With Rifampin: **Dolutegravir 50 mg BID**, Efavirenz 600 mg no change, Raltegravir 800 BID; **avoid Elvitegravir/cobicistat, Bictegravir, PIs and TAF** → use **TDF**. PI needed → Rifabutin 150 mg.
16. Pregnancy TB: RIPE + B6 (US may omit PZA → 9 months); **avoid streptomycin**. Breastfeeding safe.
17. TNF inhibitor patient: screen IGRA/CXR first; latent TB → treat ≥1 month before; active TB → hold TNFi until ≥2 months of effective therapy.

Pneumonia

1. HAP = ≥48 h after admission, not intubated; VAP = >48 h after intubation; **HCAP was retired in 2016**; hemodialysis alone is not an MDR risk.
2. Add MRSA cover if unit MRSA **>20%**; second antipseudomonal only if MDR risk/shock/unit resistance >10%. **Do not use a fluoroquinolone if local resistance >10%**.
3. **Carbapenem + Valproic acid** → VPA levels fall 60–100% → seizures (not overcome by raising VPA). **Linezolid + SSRI** → serotonin syndrome. **Daptomycin** is inactivated by surfactant → never for pneumonia.
4. HD dosing: Vanco **25 mg/kg load** after HD then 5–10 mg/kg (500–1000 mg) after each session by **pre-HD level** (15–20); Pip-tazo 2.25 g q8h + 0.75 g after HD; Cefepime 1 g q24h after HD (neurotoxicity).
5. HAP/VAP duration **7 days**; stop MRSA cover if nasal PCR and cultures negative. Ceftriaxone + levofloxacin is a CAP regimen, not VAP.
6. VAP prevention: HOB 30–45°, daily sedation interruption, toothbrushing; **routine chlorhexidine no longer advised**.
7. CAP inpatient non-severe: **Ceftriaxone 1–2 g + Azithromycin 500 mg (or Doxycycline 100 BID)**. Severe β-lactam allergy: Levofloxacin 750 / Moxifloxacin 400 (+ Aztreonam if severe CAP).
8. **QTc 527 ms → avoid macrolides and fluoroquinolones**; use Ceftriaxone + Doxycycline (+ Vanco if prior MRSA). Omadacycline has no QT effect; Lefamulin prolongs QT.
9. **Ofloxacin anaphylaxis → avoid all fluoroquinolones** (class cross-reactivity).
10. Doxycycline: separate **2 h before or 3–4 h after iron/calcium/antacids**; full glass of water, upright 30 min, sunscreen; raises INR.
11. Quinolone + calcium: levofloxacin 2 h apart; ciprofloxacin 2 h before or 6 h after; moxifloxacin 4 h before or 8 h after.
12. Severe CAP criteria: 1 major (vasopressors, ventilation) or ≥3 minor (RR ≥30, P/F ≤250, multilobar, confusion, BUN ≥20, WBC <4000, plt <100k, T <36, hypotension). Multilobar alone = 1 minor.
13. CAP duration ≥5 days and clinically stable. Aspiration pneumonia: **Ampicillin-sulbactam 3 g q6h** (CrCl 15–29: q12h); anaerobic cover only for abscess/empyema.
14. Linezolid: CBC weekly (thrombocytopenia), serotonin syndrome, tyramine, neuropathy/optic >28 days; 600 mg q12h PO = IV.

Meningitis

1. Empiric adult: **Ceftriaxone 2 g IV q12h + Vancomycin 15–20 mg/kg q8–12h** + **Ampicillin 2 g q4h if >50 y or immunocompromised** (Listeria) + Acyclovir 10 mg/kg q8h if encephalitis.
2. **Dexamethasone 0.15 mg/kg (10 mg) q6h × 2–4 days, 10–20 min before or with the first antibiotic**; stop if not pneumococcus (or H. influenzae in children).
3. Do not delay antibiotics for CT/LP. CT first only if immunocompromised, focal deficit, seizure, papilledema, altered consciousness. Blood cultures before antibiotics.
4. Bacterial CSF: WBC >1000 (neutrophils), glucose <50 (ratio ≤0.4), protein >150, high opening pressure.
5. Pneumococcus MIC <0.1 → **Penicillin G 4 MU q4h** or ampicillin; stop vanco; 10–14 days. N. meningitidis 7 days; Listeria ≥21 days; GBS 14–21.

6. Contact prophylaxis (meningococcus): **Rifampin 600 mg q12h × 4 doses**, **Cipro 500 mg × 1**, **Ceftriaxone 250 mg IM × 1** (pregnancy). Droplet isolation 24 h.
7. Ages: <1 month GBS, Listeria, E. coli; 2–50 y meningococcus, pneumococcus; >50 y add Listeria.

Endocarditis, bacteremia, osteomyelitis and aminoglycoside TDM

1. MSSA native valve: **Nafcillin/Oxacillin 2 g q4h or Cefazolin 2 g q8h × 4–6 weeks**. Prosthetic: β-lactam ≥6 weeks + **Rifampin 300 mg q8h** + **Gentamicin 2 weeks**.
2. Gentamicin **synergy 1 mg/kg q8h (3 mg/kg/day)**: peak 3–4, **trough <1**. Do not use extended-interval for synergy.
3. Gentamicin for the organism (pneumonia/sepsis): extended-interval **5–7 mg/kg q24h** on IBW (AdjBW if obese) with a **random level 6–14 h on the Hartford nomogram**; conventional 1.5–2 mg/kg q8h, peak 4–10, trough <1–2.
4. Trough timing: within 30 min before the next dose, after the 3rd–4th dose. Gentamicin ADRs: nephrotoxicity (non-oliguric, 5–10 days), ototoxicity (irreversible), NM blockade, K/Mg wasting.
5. MRSA bacteremia duration counts from the **first negative blood culture**: ≥14 days uncomplicated, 4–6 weeks complicated. ESBL with source control ~7 days.
6. **ESBL → carbapenem** (Meropenem; Ertapenem if not critically ill). Pip-tazo is inferior for ESBL bacteremia (MERINO). HD: Meropenem 500 mg–1 g q24h after HD.
7. Vancomycin target **AUC 400–600** (trough-only no longer recommended; older trough 15–20 serious). Level >20 → hold and recheck; AUC >600 → reduce 20–25%. Level <10 → verify timing first.
8. New level when: SCr changes ≥0.5 or 50%, toxicity or failure, dose change, dialysis, interacting nephrotoxins.
9. Osteomyelitis: 6–8 weeks + debridement; oral switch allowed (OVIVA). Dalbavancin 1500 mg day 1 and day 8.

Intra-abdominal, C. difficile, UTI, fungal, febrile neutropenia

1. Post-op (hospital-acquired) peritonitis: **Pip-tazo 4.5 g q6h or Meropenem**, or Cefepime + Metronidazole; **avoid quinolone + metronidazole**; source control; **4 days after source control** (STOP-IT).
2. CDI severity: severe = WBC ≥15,000 or SCr >1.5; **fulminant** = hypotension, ileus or megacolon.
3. Fulminant CDI: **Vancomycin 500 mg PO/NG q6h + Metronidazole 500 mg IV q8h**; ileus → add **rectal vanco 500 mg in 100 mL q6h**; surgery consult. Non-severe: Fidaxomicin 200 BID × 10 d (or vanco 125 QID).
4. CDI: stop the inciting antibiotic and PPI, soap-and-water hand hygiene (alcohol does not kill spores), no antimotility drugs, no routine probiotics.
5. **Candida krusei is intrinsically fluconazole-resistant** → **Micafungin 100 mg** (Caspofungin 70 → 50; Anidulafungin 200 → 100). Candidemia: remove line, eye exam, 14 days after first negative culture.
6. Pyelonephritis in pregnancy: admit, **Ceftriaxone 1–2 g** then oral step-down to 10–14 days; avoid FQ, tetracyclines, TMP-SMX (1st trimester/term), aminoglycosides; suppressive Cephalexin nightly until delivery. ESBL → Meropenem.
7. CRAB: Sulbactam-durlobactam (+ carbapenem) preferred; high-dose ampicillin-sulbactam + second agent. Colistin: nephrotoxicity + neurotoxicity (NM blockade); Polymyxin B no renal adjustment.
8. Febrile neutropenia (ANC <500, T ≥38.3 or ≥38 for 1 h): **Cefepime 2 g q8h within 60 min**; add Vanco only for MRSA colonization, instability, catheter/skin infection, pneumonia; stop at 48–72 h if cultures negative.
9. FN duration: until ANC >500 and afebrile ≥48 h, and complete the infection's own course. G-CSF not routine for established FN. Thrush: clotrimazole/nystatin; moderate–severe fluconazole.

PART E: GI AND HEPATOLOGY TRAPS

Crohn's, UC and TPN

1. **Mesalamine is not effective for Crohn's**. Mild ileal Crohn's → **Budesonide 9 mg × 8 weeks** (max ~12), taper 6 mg × 2 weeks; not for maintenance.
2. Failed azathioprine/moderate–severe → biologic: **Infliximab 5 mg/kg at weeks 0, 2, 6 then q8 weeks** (± azathioprine). Screen **TB, HBV, HCV, HIV, varicella**; no live vaccines.
3. Azathioprine: check **TPMT/NUDT15** first; CBC weekly × 4, then monthly × 3, then q3 months; LFT; pancreatitis; lymphoma/skin cancer. **Allopurinol → cut azathioprine by 75%**.
4. Perianal fistula: Metronidazole 500 TID + Cipro 500 BID × 4–8 weeks, plus biologic.
5. Fulminant UC: **Methylprednisolone 60 mg/day IV** (or hydrocortisone 100 q8h); assess at **day 3**; no response → **Infliximab or Cyclosporine 2 mg/kg/day**; VTE prophylaxis; stool C. diff.
6. **Avoid in severe colitis: loperamide, opioids, anticholinergics, NSAIDs** (toxic megacolon).
7. Rectal mesalamine at bedtime, retain as long as possible (ideally all night).
8. Crohn's vs UC: Crohn's skip lesions, transmural, mouth-to-anus, fistulas, smoking worsens; UC continuous from rectum, mucosal, bloody diarrhea, smoking 'protects'.
9. Refeeding: **Thiamine 100 mg before dextrose**, start at ~50% of goal, correct K, Mg, phosphate first.
10. Lipids: hold if **TG >400 mg/dL**, then limit to ≤1 g/kg/day. Corrected Ca = Ca + 0.8 × (4 − albumin).

GERD, PUD and H. pylori

1. H. pylori first-line: **Bismuth quadruple × 14 days** (PPI BID + bismuth + tetracycline + metronidazole). Clarithromycin triple only if known susceptibility.
2. Before UBT/stool antigen: **stop PPI ≥2 weeks**, **antibiotics and bismuth ≥4 weeks**. Test of cure **≥4 weeks after therapy**.
3. PPI after H. pylori ulcer: duodenal 4 weeks, gastric 8 weeks. Bleeding ulcer: **80 mg IV + 8 mg/h × 72 h** (or 40 mg BID).
4. NSAID need with CV risk: **naproxen + PPI**; GI risk only: celecoxib + PPI. **Ibuprofen blocks aspirin's antiplatelet effect** (take aspirin 30 min before).
5. Clopidogrel + PPI → **pantoprazole** (avoid omeprazole/esomeprazole).
6. Drugs that worsen GERD: CCBs, nitrates, anticholinergics, theophylline, benzodiazepines, bisphosphonates, NSAIDs. Long-term PPI: C. diff, fractures, B12/Mg deficiency, pneumonia.

Cirrhosis, SBP, HRS, HE, HBV, HCV

1. Child-Pugh A 5–6, **B 7–9**, C 10–15. **Avoid HCV protease inhibitors (glecaprevir, voxilaprevir) in Child B/C**.
2. SBP = ascitic **neutrophils ≥250**. **Ceftriaxone 2 g × 5–7 days + Albumin 1.5 g/kg day 1 and 1 g/kg day 3**; repeat tap at 48 h; secondary prophylaxis **Ciprofloxacin 500 mg daily** (or TMP-SMX).
3. Ascites: Na 2 g/day; **Spironolactone 100 + Furosemide 40** (ratio 100:40, max 400:160); weight loss 0.5 kg/day (1 kg with edema). Fluid restriction only if Na <125.
4. Urine Na >78 mmol/day but no weight loss = **non-adherence to diet**; urine Na low = diuretic resistance → increase doses. LVP >5 L → albumin 6–8 g per liter removed.
5. HRS-AKI: stop diuretics, NSBB, NSAIDs, nephrotoxins; **albumin 1 g/kg (max 100 g) × 2 days** then 20–40 g/day + **Terlipressin 0.85 mg q6h** (check SpO2 ≥90%: boxed respiratory failure). Day 4: no fall → stop; <30% → 1.7 mg q6h. Alternatives: NE (ICU), midodrine + octreotide.
6. HE: **Lactulose 25 mL q1–2h until ≥2 soft stools**, then titrate to 2–3/day; 5 stools/day → reduce, don't stop. Add **Rifaximin 550 BID** for recurrence. **No protein restriction.**
7. Variceal bleed: **Octreotide 50 mcg bolus + 50 mcg/h × 2–5 days**, **Ceftriaxone 1 g/day** × 7 days, EBL within 12 h, transfuse to Hgb 7–8 (not higher). Secondary prophylaxis: NSBB + EBL. Carvedilol: hold if SBP <90, AKI, HRS.
8. HBV: **Entecavir 0.5 mg (1 mg if decompensated/lamivudine-exposed) on an empty stomach**, renal dose adjust; TDF 300; TAF 25 (not in Child B/C). TDF monitoring: SCr, phosphate, urine glucose/protein, bone. **Never stop abruptly** (flare).
9. Peginterferon contraindicated in decompensated cirrhosis and severe depression. HBV pregnancy: TDF from week 28 if DNA >200,000.

10. Decompensated HCV: **Sofosbuvir/velpatasvir + ribavirin × 12 weeks** (or 24 weeks without RBV). Velpatasvir + PPI: take with food 4 h before omeprazole 20 mg (or famotidine ≤40 BID).
11. Escitalopram max 10 mg in hepatic impairment; QT with clarithromycin. ALF: NAC, transplant referral; lactulose not first-line.
12. TB in cirrhosis: fewer hepatotoxic drugs; PZA is the one to avoid in advanced cirrhosis; LFT every 1–2 weeks initially.

PART F: NEPHROLOGY AND CRITICAL CARE TRAPS

AKI and CKD

1. Contrast AKI prevention: **NS 1 mL/kg/h for 6–12 h before and after** (0.5 mL/kg/h if EF <35%); urgent: 3 mL/kg/h × 1 h before, then 1–1.5 mL/kg/h × 4–6 h after.
2. **NAC, bicarbonate, vitamin C: no benefit** (PRESERVE). Hold metformin, NSAIDs, diuretics, ACEi/ARB. SCr rises 24–48 h, peaks 3–5 days, recovers 7–14 days.
3. **FeNa = (UNa × PCr) / (PNa × UCr) × 100**: <1% prerenal; >2% ATN (muddy brown casts). On diuretics use FeUrea (<35% prerenal).
4. KDIGO stage by fold rise: 1.5–1.9× = stage 1; **2.0–2.9× = stage 2**; ≥3× or SCr ≥4 or dialysis = stage 3. Or urine output criteria.
5. Stage CKD with **CKD-EPI 2021**; dose drugs with **Cockcroft–Gault**.
6. CKD anemia: check iron first (**TSAT <20%, ferritin <100 non-dialysis / <200 HD**) → iron; ESA if Hgb <10; target 10–11.5 (never >13). Epoetin 50–100 units/kg 3×/week; darbepoetin 0.45 mcg/kg q4 weeks (actual dry weight).
7. ESA hyporesponse: iron deficiency, infection/inflammation, hyperparathyroidism, B12/folate, bleeding, aluminum, inadequate dialysis.
8. Phosphate 6: diet 800–1000 mg/day + **Sevelamer 800 mg TID with meals** (calcium-based if low Ca). Metformin max 1000 at eGFR 30–44, stop <30.
9. Iron-deficiency anemia on warfarin: classify by MCV; transfuse at Hgb <7 (<8 cardiac); ferrous sulfate 325 mg daily or every other day; separate from levothyroxine/quinolones/doxycycline.

Sepsis, shock and ICU

1. Hour-1 bundle: lactate, **blood cultures before antibiotics**, broad-spectrum antibiotics within 1 h, **30 mL/kg balanced crystalloid** for hypotension or lactate ≥4, vasopressor for MAP ≥65.
2. **Norepinephrine first** (0.05 mcg/kg/min, titrate); add **Vasopressin 0.03 units/min** at NE ≥0.25; then epinephrine. Dopamine causes more arrhythmias (SOAP II).
3. **Hydrocortisone 50 mg q6h** if NE ≥0.25 for ≥4 h. Hold antihypertensives and metformin in shock.
4. qSOFA (RR ≥22, SBP ≤100, altered mentation): a **prompt, not a diagnostic tool alone**. Albumin is not first-line fluid (only after large crystalloid volumes or for SBP).
5. Vancomycin: **load 25 mg/kg (20–35, max 3 g)**, maintenance 15 mg/kg q12h adjusted by CrCl, AUC 400–600. Post-HD level is unreliable (redistribution): use pre-HD.
6. Levetiracetam on HD: supplemental 250–500 mg after each session.
7. Hemorrhagic shock: **1:1:1** transfusion, **TXA 1 g over 10 min + 1 g over 8 h within 3 h**, calcium chloride, permissive SBP 80–90; vasopressor only as a bridge.
8. Stress ulcer prophylaxis: ventilated >48 h or coagulopathy → pantoprazole 40 mg IV. VTE: enoxaparin 40 (UFH if CrCl <30). Insulin if glucose >180, target 144–180.
9. CLABSI: 2 sets (catheter + peripheral; DTP >2 h); Vanco + Cefepime ± Micafungin if on TPN/femoral line; remove the line for S. aureus, Pseudomonas, Candida.
10. IAI: 4 days after source control. CRE: Ceftazidime-avibactam preferred (some answer keys list colistin). No routine probiotics; antifungal only with risk factors.
11. ICU PN: hold first week if low risk; early trophic ~14 kcal/kg; **peripheral PN ≤900 mOsm/L** (dextrose 1% ≈ 50 mOsm, amino acid 1% ≈ 100, electrolytes 2 mOsm/mEq).
12. PN protein: CKD not on dialysis 0.6–0.8 g/kg; **HD 1.2–1.5**; CRRT up to 2.5. Insulin in bag 0.05–0.1 unit/g dextrose.
13. Acid–base: AG = Na − (Cl + HCO3), normal 12; **Winter: expected PaCO2 = 1.5 × HCO3 + 8 ± 2**. Metabolic alkalosis with urine Cl <25 → saline-responsive.
14. Urosepsis relapse within 14 days → look for an undrained source/abscess, not just change antibiotics.

PART G: RESPIRATORY, NEURO/PSYCH AND RHEUM TRAPS

Asthma and COPD

1. **Never SABA-only** in adults. GINA track 1: as-needed **budesonide-formoterol**; MART step 3 low dose 1 inhalation BID, step 4 medium 2 BID + PRN, max 12/day.
2. High SABA use (≥3 canisters/year) = risk of severe exacerbation/death. Check technique and adherence **before stepping up**.
3. Moderate exacerbation: SpO2 93–95%, SABA, **prednisolone 40–50 mg × 5–7 days**; Mg 2 g IV if severe.
4. COPD GOLD grade by FEV1 (GOLD 2 = 50–79%); group: **A** (0–1 moderate, mMRC 0–1/CAT <10) → one bronchodilator; **B** → LABA + LAMA; **E** (≥2 moderate or ≥1 hospitalization) → LABA + LAMA, add ICS if eos ≥300.
5. COPD exacerbation: **O2 88–92%**; NIV if pH ≤7.35 and PaCO2 ≥45; **prednisone 40 mg × 5 days**; antibiotics if 3 cardinal symptoms or 2 incl. purulence, 5–7 days.
6. COPD + HF → β1-selective blocker is fine. ACEi cough → losartan. COPD depression with weight loss/insomnia → mirtazapine.
7. Inhaler traps: MDI prime 4 sprays (new or unused >14 days), wait 1 min between puffs; **Diskus/DPI: no spacer, no shaking, breathe fast and deep**; rinse after ICS; HandiHaler capsule not swallowed; Respimat slow inhalation.
8. Smoking cessation: **Varenicline 0.5 mg daily × 3 d → 0.5 BID × 4 d → 1 mg BID × 12 weeks** (quit date day 8); **Bupropion SR 150 daily × 3 d → BID**, contraindicated with seizures, eating disorders, MAOI.

Neurology and psychiatry

1. tPA exclusions to recite: ICH history, **BP >185/110 not controlled**, platelets <100k, **INR >1.7**, DOAC within 48 h, glucose <50, head/spinal trauma or stroke within 3 months, major surgery within 14 days (relative). Unknown onset time = cannot decide.
2. **Alteplase 0.9 mg/kg (max 90 mg): 10% bolus over 1 min, rest over 60 min**; Tenecteplase 0.25 mg/kg (max 25). Window 4.5 h; thrombectomy up to 24 h.
3. After tPA: **no antiplatelet/anticoagulant for 24 h**, BP <180/105. No tPA: permissive BP up to 220/120, aspirin 160–325 within 24–48 h.
4. Minor stroke (NIHSS ≤3) or high-risk TIA: **Clopidogrel 300 → 75 + Aspirin × 21 days**, then single agent. Atorvastatin 80.
5. Neurogenic shock: NE, MAP 85–90 for 7 days. Bradycardia: atropine 1 mg (max 3) → dopamine/epinephrine/pacing.
6. **Carbamazepine + warfarin** (APS): induction lowers INR → **Levetiracetam 500 mg BID**; if CBZ kept, raise warfarin and monitor INR closely. Communicate with SBAR.
7. Duloxetine 30 → 60 mg for neuropathy + depression: BP, Na (SIADH), LFT; avoid CrCl <30. With linezolid → serotonin syndrome.
8. Antidepressant trial 6–8 weeks at an adequate dose; continue 6–12 months after remission. Insomnia: **CBT-I first**; zolpidem 5 mg in women/elderly (Beers).

Rheumatology

1. RA: **MTX 15 mg once WEEKLY** → 20–25 mg + **folic acid 1 mg daily**; switch to SC if inadequate; then add TNFi. **Daily MTX dosing is a fatal error.**
2. MTX labs: CBC, LFT, SCr q2–4 weeks × 3 months, then q8–12 weeks, then q12 weeks. Avoid TMP-SMX, alcohol, NSAID excess.
3. Before any biologic: **TB (IGRA/CXR), HBV, HCV, HIV**, pregnancy, update vaccines (no live vaccines on therapy).
4. Pregnancy planning: **stop MTX ≥3 months before conception** (some labels 6 months); leflunomide → cholestyramine washout; HCQ, sulfasalazine, certolizumab/etanercept are OK; infant BCG delayed 6 months if TNFi in late pregnancy.
5. Infliximab RA 3 mg/kg weeks 0, 2, 6 then q8 weeks; adalimumab 40 mg q2 weeks; rituximab 1000 mg days 1 and 15. JAK inhibitor boxed warning: age ≥50 with CV risk, VTE, malignancy.

6. Assess response at 3 months (≥50% improvement), target remission/low activity by 6 months. HCQ eye exam baseline and yearly after 5 years.
7. Osteoporosis fracture on alendronate → zoledronic acid, **denosumab (no drug holiday; follow with a bisphosphonate)** or teriparatide. Romosozumab not within 1 year of MI/stroke.
8. Med review traps: dual RAAS blockade, aspirin + warfarin in stable CAD + AF (OAC alone), HCTZ worsening incontinence, ranitidine withdrawn, **calcium citrate with acid suppression**.

PART H: PATIENT SAFETY AND COMMUNICATION STATIONS

1. **Treat first, talk second**: anaphylaxis → stop the drug, **Epinephrine 0.01 mg/kg IM (0.3–0.5 mg of 1 mg/mL) anterolateral thigh**, repeat q5–15 min, fluids 1–2 L, supine with legs raised. Adjuncts (diphenhydramine, famotidine, salbutamol) **never delay epinephrine**.
2. On a β-blocker and refractory: **Glucagon 1–5 mg IV**. Observe 4–6 h (biphasic up to 72 h); tryptase within 1–2 h.
3. Angry patient: introduce yourself, sit at eye level, listen without interrupting, empathize, **apologize and take responsibility, no blaming colleagues**, explain honestly, give a contact.
4. Report: **hospital incident report + ADR report to SFDA (National Pharmacovigilance Center)**; document; add allergy alert and band; RCA; EHR hard-stop; epinephrine auto-injector and training at discharge.
5. Medication error with harm (warfarin INR 6 + epistaxis): manage bleeding, then disclose; prevention = **two-person check, read-back of verbal orders, written dose calendar, teach-back**; consider a DOAC if no mechanical valve/APS.
6. Difficult/frightened patient with drug toxicity (INH): acknowledge fear, avoid jargon, explain the injury is usually reversible, stop the drug, give red flags, report the ADR.
7. Counseling stations are graded on structure: **open** (introduce, confirm identity, ask what they know) → content → **check interactions** → **teach-back** → close with follow-up.

PART I: CALCULATIONS THAT APPEARED IN THE EXAMS

Calculation	Formula	Worked example / trap
CrCl (Cockcroft–Gault)	(140 − age) × weight / (72 × SCr mg/dL), × 0.85 if female	SCr µmol/L ÷ 88.4 = mg/dL (109 µmol/L = 1.23 mg/dL). DOAC labels used actual body weight
IBW	Men 50 + 0.91 × (cm − 152.4); women 45.5 + 0.91 × (cm − 152.4)	Or 50 / 45.5 + 2.3 kg per inch over 5 ft. AdjBW = IBW + 0.4 × (TBW − IBW)
BMI	kg / m²	86 kg, 175 cm = 28 (overweight)
Enoxaparin	1 mg/kg q12h or 1.5 mg/kg q24h; q24h if CrCl <30	62.4 kg → 60 mg q12h (round to syringe)
UFH (VTE)	80 units/kg bolus + 18 units/kg/h	ACS: 60 units/kg (max 4000) + 12 units/kg/h; PCI 70–100 units/kg
Vancomycin	Load 25 mg/kg actual weight (max 3 g); maintenance 15 mg/kg q8–12h by CrCl	HD: 25 mg/kg then 500–1000 mg after each HD; pre-HD level
Gentamicin	Extended 5–7 mg/kg q24h (IBW/AdjBW); synergy 1 mg/kg q8h	IBW 70 × 7 = 490 mg q24h; random level 6–14 h → Hartford
FeNa	(UNa × PCr) / (PNa × UCr) × 100	<1% prerenal; 3.4% = ATN
Anion gap / Winter	Na − (Cl + HCO3); PaCO2 = 1.5 × HCO3 + 8 ± 2	AG 20 = HAGMA; check delta ratio
Corrected Ca	Ca + 0.8 × (4 − albumin)	Albumin 2: Ca 7.6 → 9.2
Insulin	TDD 0.4–0.6 units/kg; ICR 500/TDD; CF 1800/TDD	TDD 50 → ICR 10 g/unit, CF 36 mg/dL/unit
TPN energy	Total kcal = kg × kcal/kg; protein kcal = g × 4; dextrose kcal = g × 3.4; lipid 20% = 2 kcal/mL (10 kcal/g)	80 kg × 25 = 2000 kcal; dextrose 4 g/kg = 320 g × 3.4 = 1088 kcal
GIR	dextrose g × 1000 / kg / 1440	320 g / 80 kg → 2.8 mg/kg/min (limit 4–5)

Calculation	Formula	Worked example / trap
Dextrose from a bag	g = volume (mL) × % / 100 (or rate × 24 × %)	D70 500 mL = 350 g
K phosphate	mL = mmol P / 3; K (mEq) = mL × 4.4	15 mmol P = 5 mL = 22 mEq K
Peripheral PN osmolarity	dextrose % × 50 + AA % × 100 + mEq electrolytes × 2	Must be ≤900 mOsm/L
Furosemide conversion	IV 1 mg = PO 2 mg (≈ torsemide 0.5, bumetanide 0.025)	Home 20 mg PO = 10 mg IV; ADHF give 1–2.5× home dose IV
CHA2DS2-VASc / HAS-BLED	See Part B	69 y + HTN + DM = 3; +MI = 4; female +1
Child-Pugh	Bilirubin, albumin, INR, ascites, HE (1–3 points each)	A 5–6, B 7–9, C 10–15

PART J: COUNSELING CHECKLISTS EXAMINERS LISTEN FOR

Drug	Must-say points
Warfarin	Same time daily, consistent vitamin K (greens), no NSAIDs/aspirin/herbals, report new antibiotics, bleeding signs (black stool, red urine, severe headache), never double a missed dose, INR schedule, carry a card, teratogenic
DOACs	Twice-daily adherence (short half-life), Rivaroxaban 15/20 mg with the evening meal, missed-dose rules, no INR but yearly CBC/SCr/LFT, interactions (rifampin, carbamazepine, azoles)
Enoxaparin	SC abdomen, rotate sites, **do not expel the air bubble or rub**, bruising common, bleeding signs, stop 24 h before delivery/neuraxial, safe in breastfeeding
SL nitroglycerin	Sit down, 0.4 mg SL q5 min up to 3; **call EMS if pain not better 5 min after the first dose**; keep in original glass bottle; no PDE5 inhibitors
ARNI / ACEi / MRA / SGLT2i	36-h washout, angioedema, dizziness, hyperkalemia (avoid salt substitutes), gynecomastia, genital hygiene, hold SGLT2i on sick days, daily weights (call if +1 kg/day or +2 kg/week)
Amiodarone	With food, sunscreen, report cough/dyspnea, vision change, thyroid symptoms; many interactions
Insulin	Storage, 28-day in-use, site rotation, timing with meals, 15-15 rule, glucagon, sick-day rules, never skip basal
Levothyroxine	Empty stomach 30–60 min before breakfast, 4 h apart from Ca/Fe/PPI, same brand, TSH 6–8 weeks
Methotrexate	**Once weekly**, folic acid, no alcohol, avoid TMP-SMX, contraception and stop 3 months before conception, report mouth sores/fever/cough
Alendronate	Empty stomach with full glass of plain water, upright 30 min, no food/drugs for 30 min, calcium + vitamin D, dental check
TB drugs	Empty stomach, DOT, orange fluids, vision change (ethambutol) and jaundice → stop and call, avoid alcohol, OCs fail with rifampin
Doxycycline	Full glass of water, upright 30 min, sunscreen, separate from iron/calcium/antacids
Inhalers	Device-specific technique, rinse after ICS, spacer with MDI, prime MDI, no spacer with DPI, controller daily even when well
Smoking cessation	Quit date, varenicline nausea and vivid dreams, bupropion seizure risk, NRT combination works best, 5 A's

PART K: TOP 40 LAST-MINUTE TRAPS

1. Aspirin loading is 162–325 mg chewed, not 81 mg.
2. Flecainide after MI = contraindicated (CAST).
3. Diltiazem/verapamil in HFrEF = harmful; amlodipine in HFrEF = stop (no benefit).
4. AF + PCI = DOAC + clopidogrel; aspirin ≤1 week.
5. Apixaban 2.5 mg needs 2 of 3 (age ≥80, ≤60 kg, SCr ≥1.5).
6. Warfarin → apixaban at INR <2; → rivaroxaban at INR <3.
7. Mechanical valve and APS = warfarin, never DOAC.
8. ACEi → ARNI needs a 36-hour washout.
9. Spironolactone only if eGFR >30 and K <5.0; check K at 1 week, 1 month, 6 months.
10. Digoxin, ivabradine, vericiguat and loops do not lower mortality.
11. Simvastatin max 20 mg with amiodarone; amiodarone cuts warfarin and digoxin doses by 30–50%.
12. Oxygen in ACS only if SaO2 <90%; in COPD target 88–92%.
13. Contrast AKI: saline yes; NAC and bicarbonate no.
14. Minor warfarin bleed: oral vitamin K; IV vitamin K 10 mg + PCC is only for major bleeding.
15. Early high INR during warfarin start is not full anticoagulation: keep heparin ≥5 days and INR ≥2 × 24 h.
16. Carbamazepine/rifampin lower INR; amiodarone/fluoxetine/antibiotics raise it.
17. Anti-Xa is a peak level 4–6 h after the 3rd–4th dose; treatment target 0.6–1.0.
18. DKA: check K before insulin; K <3.3 = no insulin.
19. DKA: add dextrose at glucose 200, keep insulin running; overlap SC 1–2 h.
20. Bicarbonate in DKA only if pH <7.0.
21. Thyroid storm: PTU before iodine (≥1 h); acetaminophen not aspirin.
22. Myxedema coma: steroid before levothyroxine.
23. ICR 500/TDD, CF 1800/TDD, T1DM 50/50 basal/bolus.
24. TB liver stop: ≥3× ULN with symptoms or ≥5× without; rechallenge RIF → INH → PZA.
25. INH hepatitis in latent TB: do not rechallenge; 4 months rifampin.
26. Ethambutol = eyes; PZA = gout; INH = neuropathy (B6); RIF = orange fluids and interactions.
27. Voriconazole + rifampin = contraindicated → amphotericin B.
28. HIV-TB: Dolutegravir 50 mg BID, TDF not TAF, no cobicistat; ART 2 weeks (CD4 <50) or 8 weeks.
29. Carbapenem + valproate = seizures; linezolid + SSRI = serotonin syndrome; daptomycin not for pneumonia.
30. QTc >500: no macrolide or quinolone; use doxycycline.
31. Meningitis: dexamethasone before or with the first antibiotic; ampicillin if >50 years.
32. Gentamicin synergy trough <1; extended-interval uses the Hartford nomogram.
33. Candida krusei = fluconazole-resistant → echinocandin.
34. Fulminant CDI = vanco 500 PO q6h + IV metronidazole (+ rectal vanco if ileus).
35. Mesalamine does not work in Crohn's; budesonide is not maintenance.
36. Severe colitis: no loperamide, opioids, anticholinergics or NSAIDs.
37. H. pylori testing: off PPI 2 weeks, off antibiotics/bismuth 4 weeks.
38. SBP: ceftriaxone + albumin 1.5 g/kg day 1 and 1 g/kg day 3; HE: no protein restriction.
39. MTX is weekly, with folic acid; stop 3 months before conception.
40. Anaphylaxis: IM epinephrine first; then apologize, report to SFDA, add the allergy alert.

Note: numbers follow the answer keys in your compilations and current guidelines (ACC/AHA, ADA, IDSA, KDIGO, AASLD, GINA/GOLD). If your course slides give a different number, use the course number in the exam.